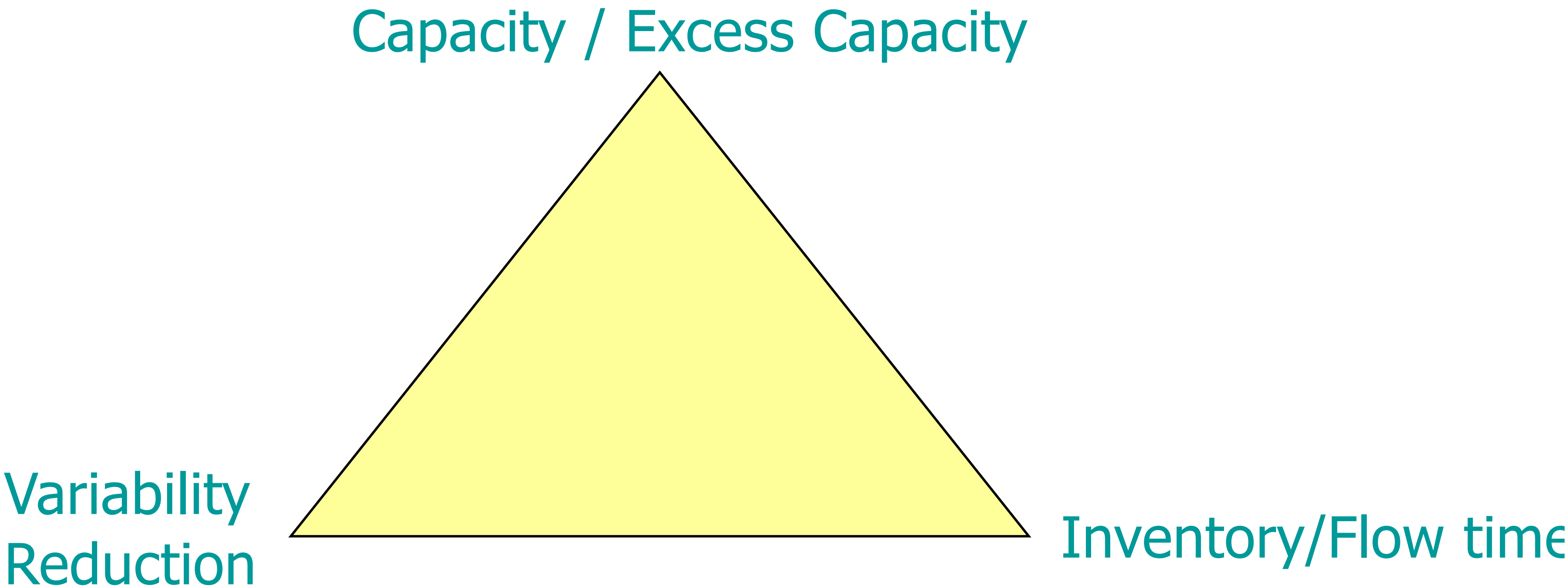
Supply Chain Management

Inventory and Risk

MM BASC 523

April-May 2026

OM Triangle



Capacity, inventory and variability reduction (information) are substitutes in running operations…

Managing Operational Risk

How much to buy?

- Demand = 1,2,3,… or 10. Equally Likely
- Revenue = $3 per unit
- Cost = $1 per unit

HOW MUCH TO BUY?

8	4	10	2	9
2	3	5	3	1
7	9	1	8	10

What is your forecast for next week's demand?

Forecast?

How do you Forecast?

- What do you need?
- How will you do it?

Example

Sales for the last 12 weeks are

2 , 5, 7, 7, 7, 8, 3, 2, 7, 4, 5, 3

Empirical distribution

- Village idiot's solution (assuming u see real demand)

  2 , 5, 7, 7, 7, 8, 3, 2, 7, 4, 5, 3

Distribution:

- 4, 8 with probability = 1/12 each.
- 2, 3, 5 with probability = 2/12 each.
- 7 with prob = 4/12

How will a real demand forecast look like?

How will a real demand forecast look like?

- Will be a probability distribution
- Will have relationship to
  - Price
  - Seasonality
  - Trends
  - Category effects….
- The simplicity of a forecast is important
  - Compact representations that does not compromise the effectiveness…(Tricky)

Today

- Risk: An example.
  - **How much inventory?**
- Same framework for capacity.

- Similar framework for lots of decisions that one would make to respond to a risky outcome.

Newsvendor Model

Risk and Inventory

- The idea of Marginal analysis

How much to buy?

- Demand = 1,2,3,… or 10 equally likely
- Revenue = \$3 per unit
- Cost = \$1 per unit

HOW MUCH TO BUY?

How much to buy?

- Demand = 1,2,3,… or 10 equal likelihood.
- Revenue = \$3
- Cost = \$1

What is the cost of having **ONE** too many units

Overage cost = \$O per unit = \$1

What is the cost of having **ONE** too few units

Underage cost = \$U per unit = \$2

Marginal analysis of inventory under Risk

Illustration

- Demand = 1,2,3,… or 10 equal likelihood.
- Revenue = $3
- Cost = $1

Say I have Q = 2 units with me and I am evaluating the marginal value of the next unit (i.e. the 3rd unit)

Marginal value of the third unit (on hand Q = 2)

– **Potential Benefit:**

- I will be able to sell this unit if demand exceeds **Q = 2.**
- **In this case I make a profit r-c = $2**
- **What is the likelihood this will happen?**

Marginal value of the third unit (on hand Q = 2)

- **Potential Benefit:**
  - I may be able to sell this unit if demand exceeds **Q = 2.**
  - **In this case I make a profit r-c = $2**
  - **What is the likelihood this will happen?**
    - **Demand must be 3 or 4 or 5 ….etc. 10 so it a 80% chance.**

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- **Potential Loss**
  - I may be stuck with it (incur the cost of a $1).
  - This happens if demand is lesser than or equal to **Q = 2.**
  - **This will happen with a 20% chance.**

Marginal value of the third unit (on hand Q = 2)

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– Potential Loss
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  - This happens if demand is lesser than or equal to Q = 2.
  - This will happen with a 20% chance.

Marginal Value of the third unit equals

$2* 0.8 – $1* 0.2 = $1.4 greater than zero!

So you should buy the third unit!

- What is optimal number of units?

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  - **If you keep doing this you will realize you will stop at 7!**

So you should buy the third unit!

- What is optimal number of units?
  - **If you keep doing this you will realize you will stop at 7! That is 70% of the demand.**
  - **What is special about 7 or the 70%?**

Recall

What is the cost of having **ONE** too many units

Overage cost = \$O per unit = \$1

What is the cost of having **ONE** too few units

Underage cost = \$U per unit = \$2

$$\frac{U}{U+O} = 70\% \text{ rounded up}$$

Marginal Value

- If you have Q units and you want to get to Q+e units (e is small), what is the marginal value of this e units?

- Marginal Value =

$$r * \text{Probability } (D>Q) - c$$

- Set MV = 0!

$$\frac{U}{U+O}$$

The per unit Underage cost
That is the Per unit cost of having too little.

The per unit Overage cost
That is the Per unit cost of having too much.

Service Level

$$\frac{U}{U+O}$$

The per unit Underage cost
That is the Per unit cost of having too little.

The per unit Overage cost
That is the Per unit cost of having too much.

Example

- Kellogg Cereal, Vendor managed Inventory from Monterey Mexico.
  -

Data

<u>Example (Kellogg Cereal in Mexico)</u>

- **Wholesale price = $2.7**
- **Retail Margin = 20%**
  - **r = $3.24**
- **Weekly replenishment to Walmart from DC.**

Service level at Walmart!

<u>Example (Kellogg Cereal)…</u>

Optimal service level

$$\frac{\mathbf{r-c}}{\mathbf{r}} = \frac{\mathbf{3.24 - 2.7}}{3.24}$$

$$= \mathbf{17\%}$$

Good will cost
(the cost of a lost sale)

<u>Example (Kellogg Cereal)…</u>

Optimal service level

$$\frac{r - c}{r} = \frac{3.24 - 2.7}{3.24}$$

$$= 17\%$$

What is happening? Is this a reasonable service level?

Service level Insight

The optimal service level in many settings can be given by the following trade-off

- What is the cost of having **ONE** too many units
  - **Overage cost = $O per unit**
- What is the cost of having **ONE** too few units
  - **Underage cost = $U per unit**

Intuition behind optimal Service Level expression

$$\text{Optimal Service Level} = \frac{U}{U+O}$$

The per unit Underage cost
That is the Per unit cost of having too little inventory

The per unit Overage cost
That is the Per unit cost of having too much inventory

Optimal service level:

Reconciling with what we just did in the newspaper example:

What is the Overage cost: **O** = \$c

What is the Underage cost: **U** = \$ (r-c)

$$\frac{\mathbf{U}}{\mathbf{U+O}} = \frac{\mathbf{r - c}}{\mathbf{r - c + c}} = \frac{\mathbf{r - c}}{\mathbf{r}}$$

Goodwill cost (g)

- What happens if you add a good will cost to the calculation?

What is the Overage cost: **O** = $c

What is the Underage cost: **U** = $(r+g-c)

$$\frac{\mathbf{U}}{\mathbf{U+O}} = \frac{\mathbf{r+g-c}}{\mathbf{r+g-c+c}} = \frac{\mathbf{r+g-c}}{\mathbf{r+g}}$$

Good will cost
(the cost of a lost sale)

<u>Example (Kellogg Cereal)…</u>

If Optimal service level = 90%

$$\frac{r+g-c}{r+g}=\frac{3.24+g-2.7}{3.24+g}=0.9$$

Goodwill = g = $23.75

Calculating Order Quantity from a Forecast

(Example 1)

Demand for a new I-pad in a store is Normally distributed with a mean (average) of 10,000 units and a standard deviation of 1000 units.

Retail price: r = $600.

The wholesale price the store paid Apple = $300 per unit.

g= $200 per unit.

HOW MUCH SHOULD THE STORE ORDER?

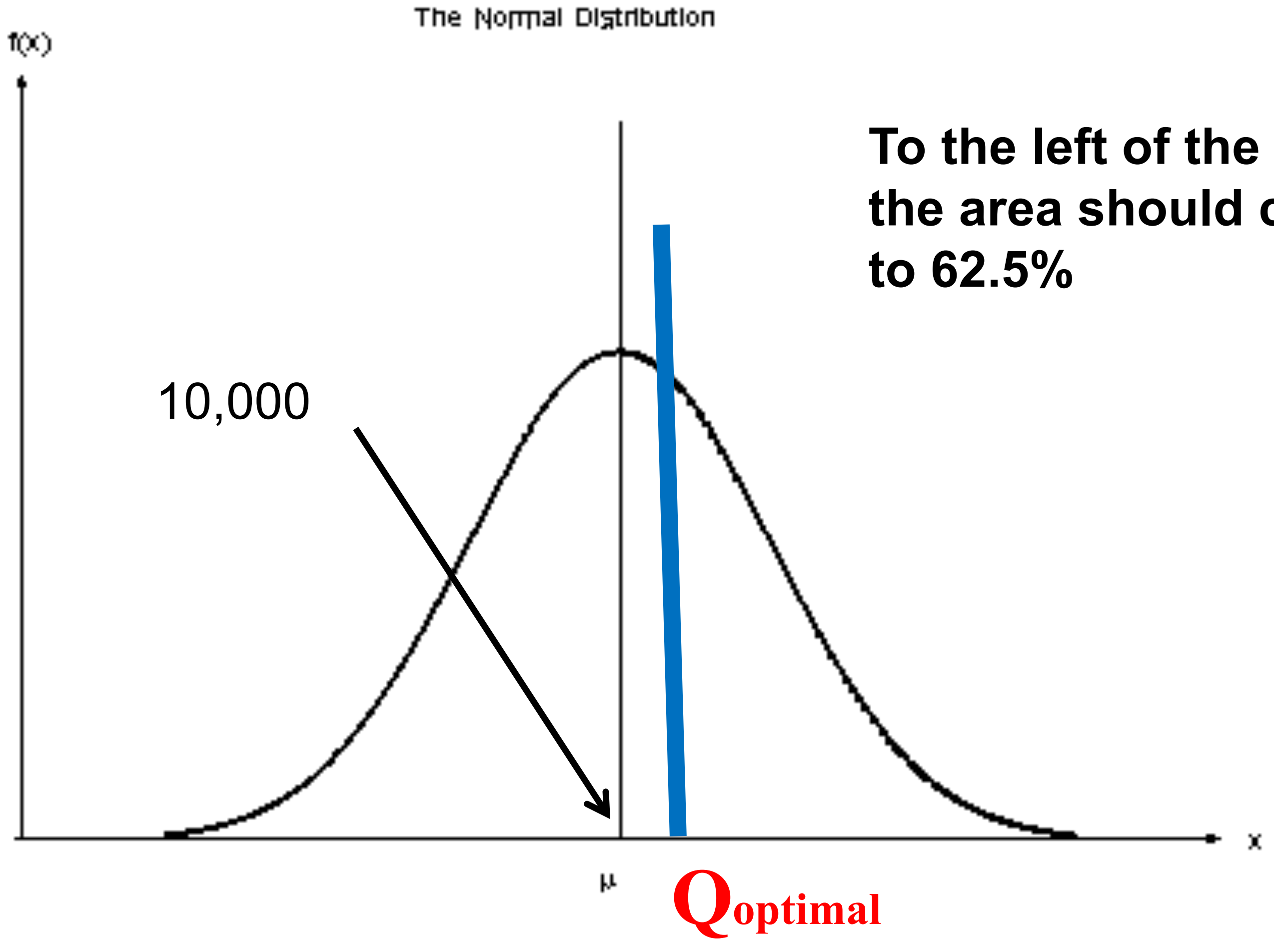
Example 1

- Optimal Service level: **(r +g - c) ÷ (r+ g)**

= (800 - 300) ÷ (800)

= 5 ÷ 8 = 0.625, i.e. **62.5%** service level.

(Example 1: Ipads)



To the left of the blue line, the area should correspond to 62.5%

Demand Forecast for tomorrow' s newspaper sales is **N (10000, 1000)**

Normal Distribution
Optimal Q

Q = Mean demand + **z** * Standard deviation of demand

Comes from the optimal service level

Normal Distribution
Optimal I-pads

Q = 10,000 + **0.32** * 1000

= **10320 ipods.**

The z value corresponding to 62.5%

(NORMSINV(0.625) IN EXCEL)

Calculating Order Quantity from a Forecast

(Example 2)

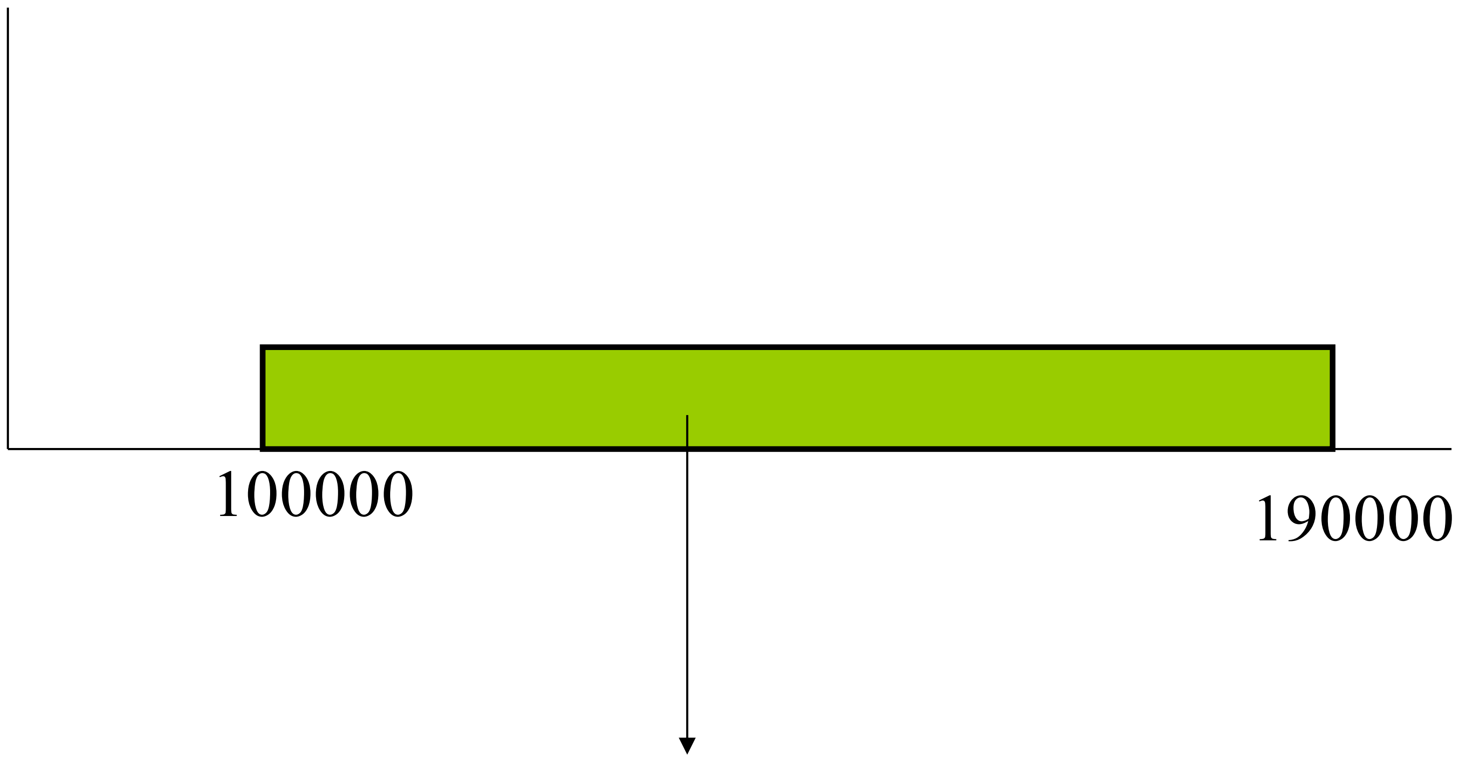
You have a product for which you need to calculate the optimal inventory level.

Your Marketing manager tells you that the optimal service level is 75%.

Your statistician friend has looked at past sales data, worked her magic and comes up with a forecast that **demand is uniformly distributed between 100000 and 190000**.

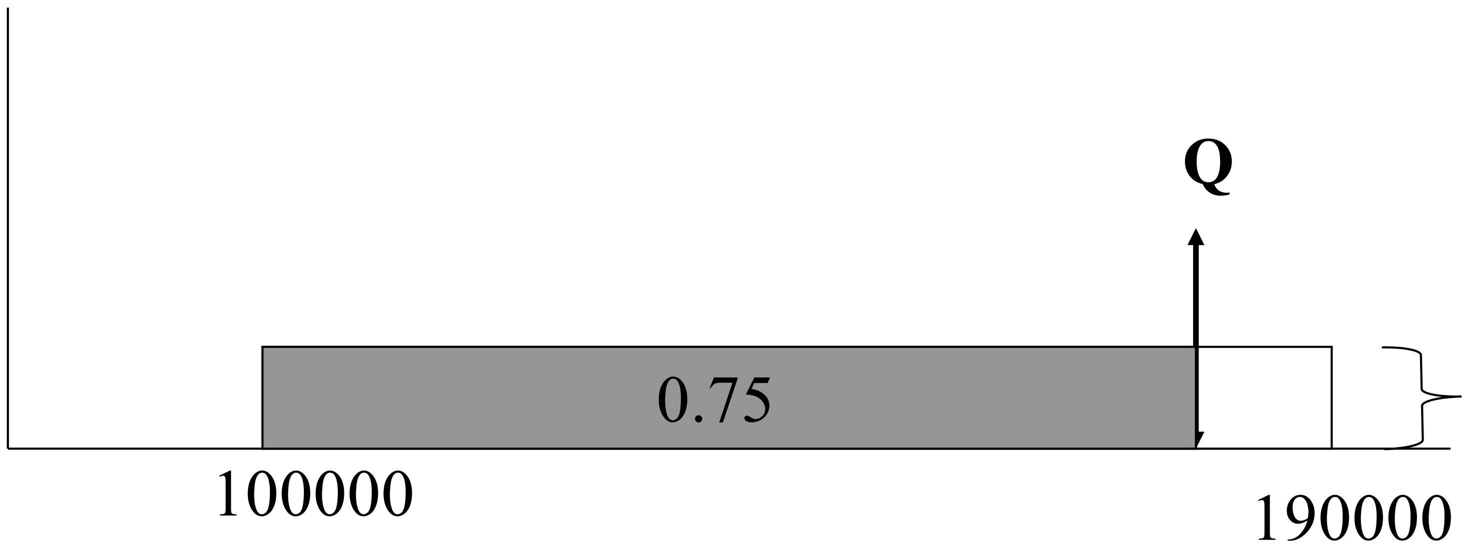
How do you decide Q?

Uniform distribution

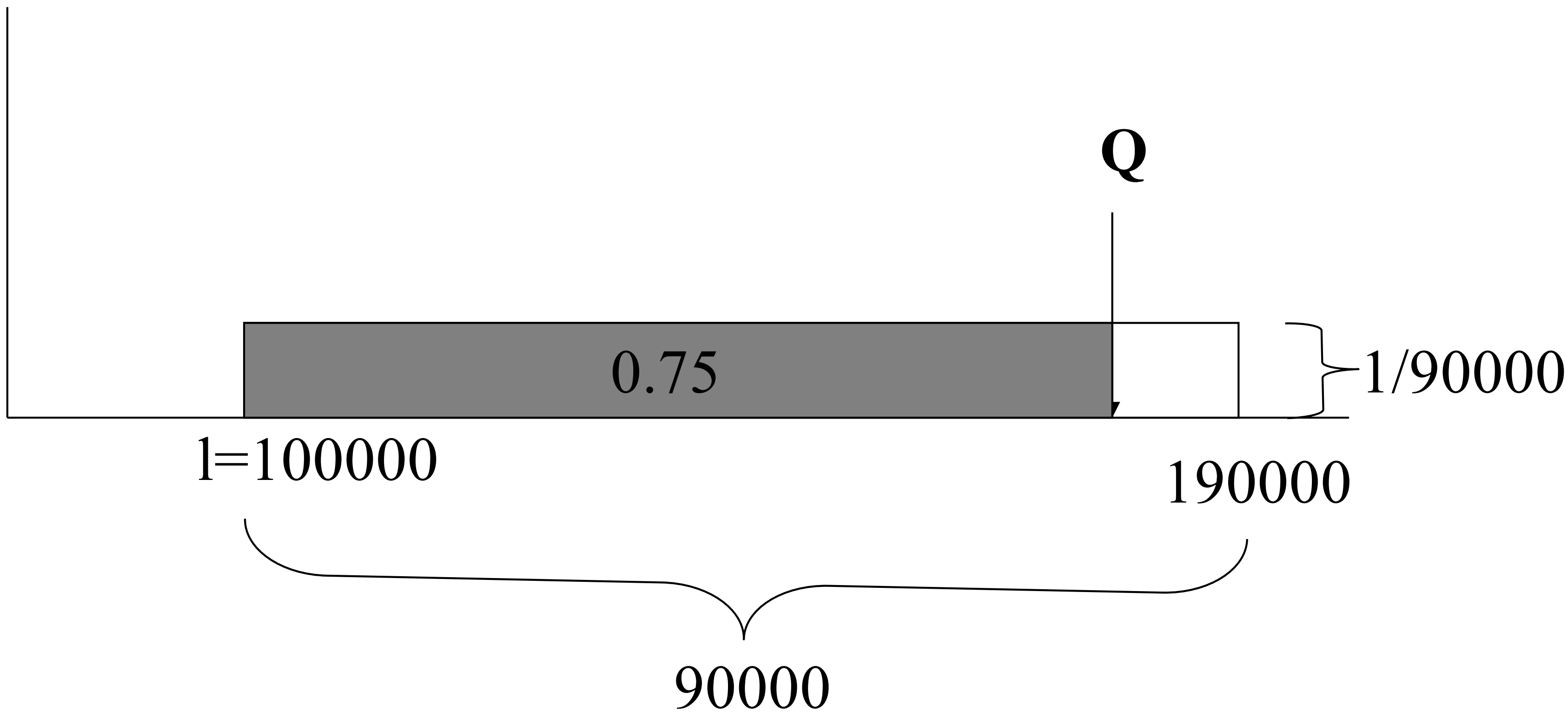


The Total Area (i.e. the green rectangle) is 1.

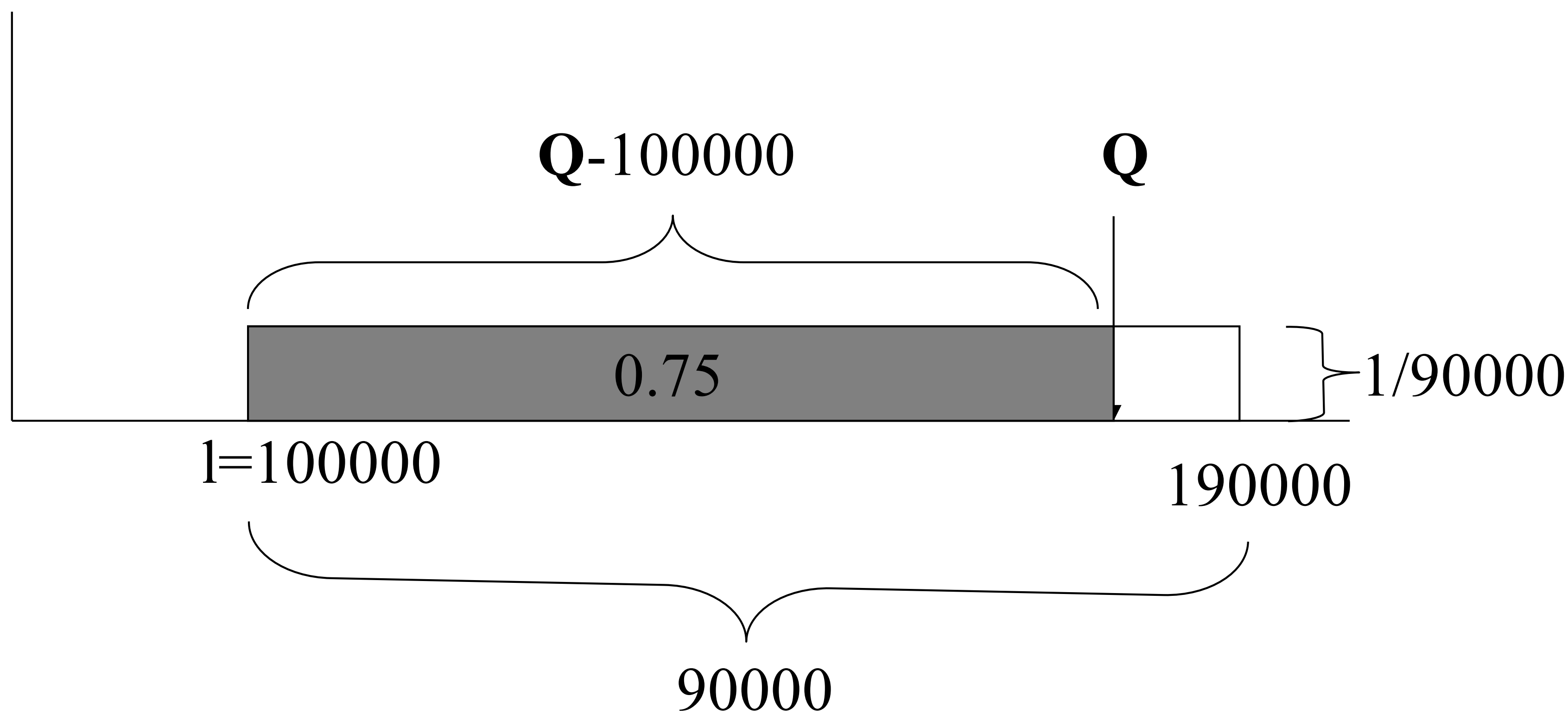
Uniform distribution
(75% service level)



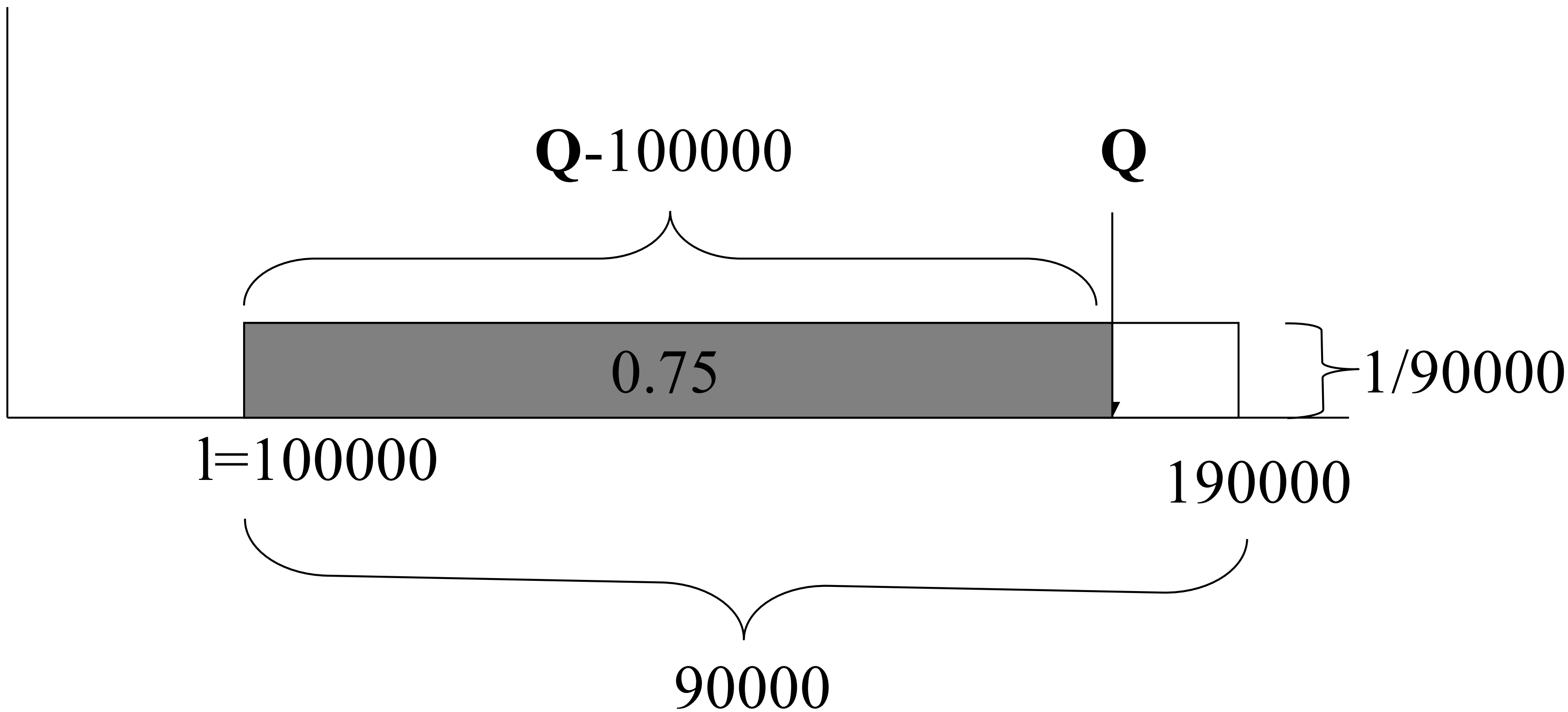
Uniform distribution



Uniform distribution



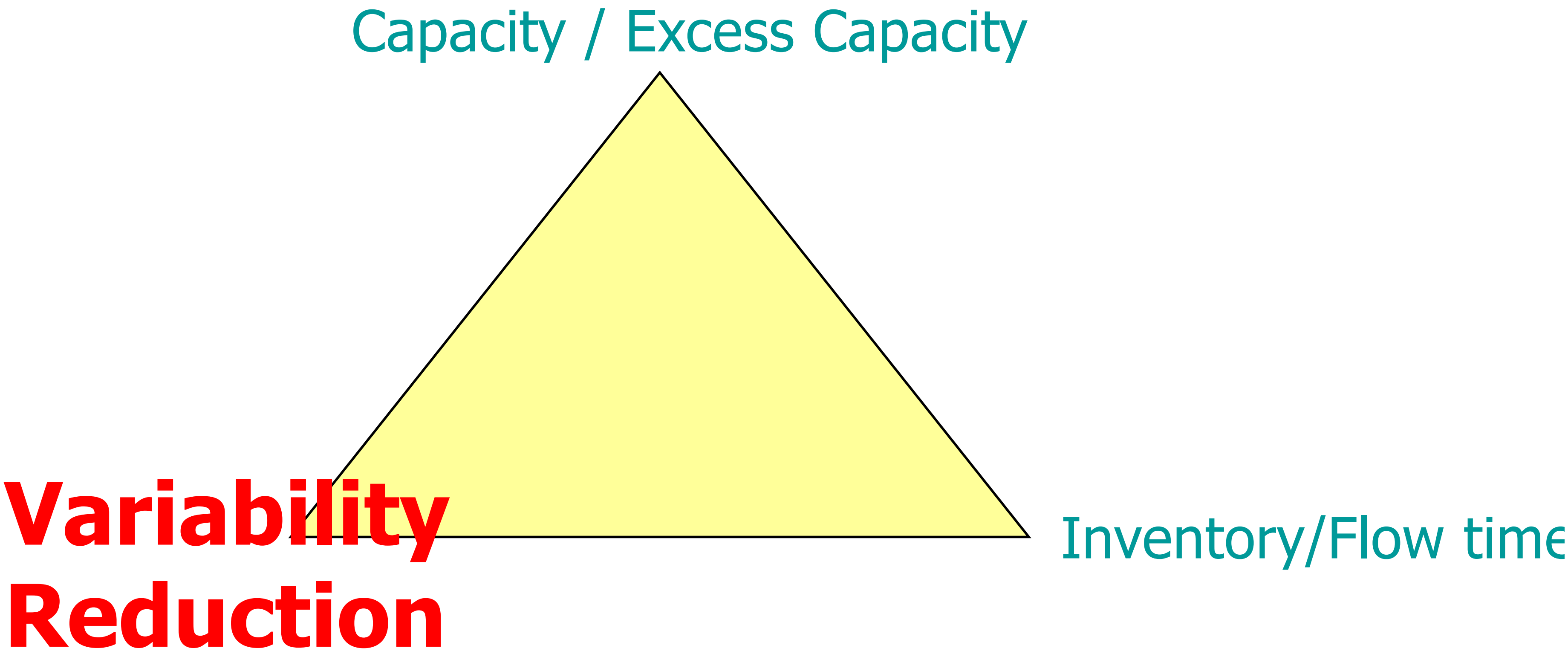
Uniform distribution



(Q-100000)*1/90000=0.75

Q=167500

OM Triangle



Capacity, inventory and variability reduction (information) are substitutes in running operations…

Warehouse and DCs

Simple Example

Two retail outlets, each with a demand forecast of (say) N(10000, 2000) (i.e., average demand is 10000 and standard deviation is 2000): Independent demands:

Each store requires a service level that corresponds to 99.99% = 3 standard deviations.

Inventory level at each retailer will then be

= 10,000+3*2000 = 16000 units.

Total inventory = 32,000 units.

Warehouse and DCs

Simple Example continued…

Let us combine the forecasts of the 2 outlets and keep the inventory for them in a centralized warehouse.

New Forecast:

Average demand = 10,000+10,000 = 20,000

Standard Deviation = $\sqrt{(2000)^2 + (2000)^2} = 2000 * \sqrt{2}$

= 2829

Warehouse and DCs

Simple Example continued…

To ensure a 99.99 service level with the combined forecast

$$Q = 20{,}000 + 3^* \ 2829$$

$$= 28487 \text{ units}$$

You save 32000 - 28487 = 3513 units and achieve the same service level!!! A direct consequence of Risk Pooling.

Combining Random Variables

Think of forecasts of two products:

- Forecast of Blue Sweaters.
  - Normal: Ave:1000 and Standard deviation: 200
- Forecast of Green Sweater.
  - Normal: Ave: 2000 and Standard Deviation: 350

Assume the demands are independent.

Joint Forecast

- When you combine two random variables
  - Their average add up.
  - Their Variances add up if they are independent.
  - They continue to be Normally distributed.
- Combined average of green sweater and blue sweater = 1000 + 2000 = 3000 units.

Combined Standard deviation

Variances add up (indep demands)

- Variance of Green sweater: 350*350
- Variance of Blue sweater: 200*200
- Combined Variance = 122500 + 40000

  = 162500
- Combined Standard deviation =
  - Square Root (162500) = 403
  - This is less than 350 + 200!

Forecasts

- Combined Forecast:
  – Normal with an average of 3000 and a standard deviation 403

Individual:

- Green: Normal: Avg: 2000: Std. Dev: 350
- Blue: Normal: Avg: 1000: Std. Dev: 200

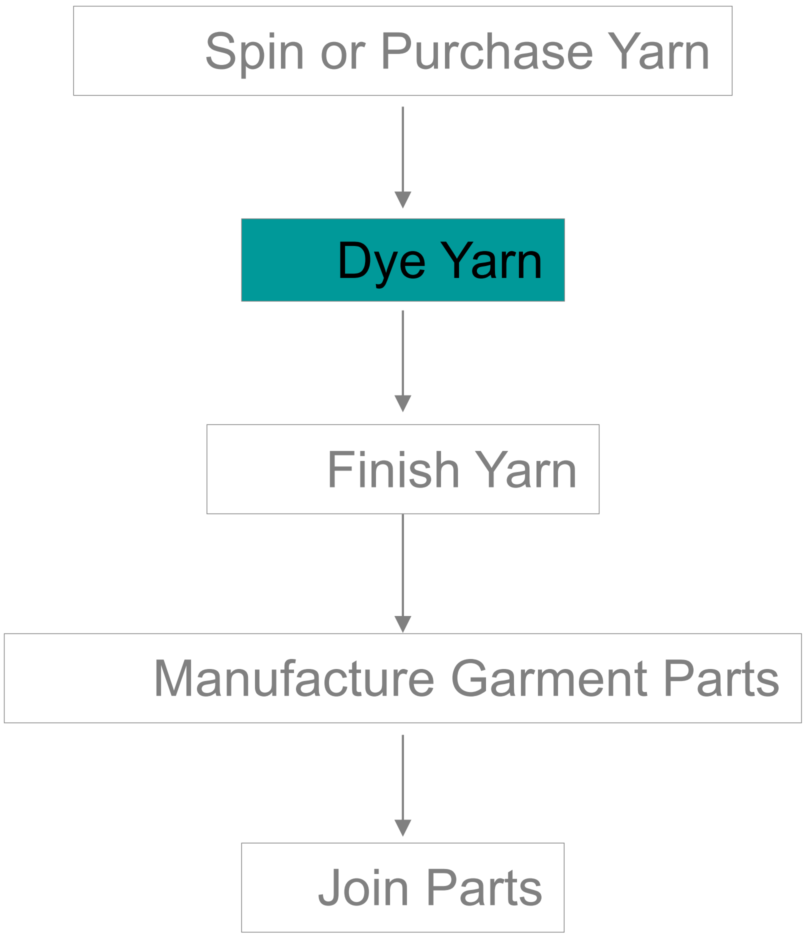
Why is this useful?

Ideas!

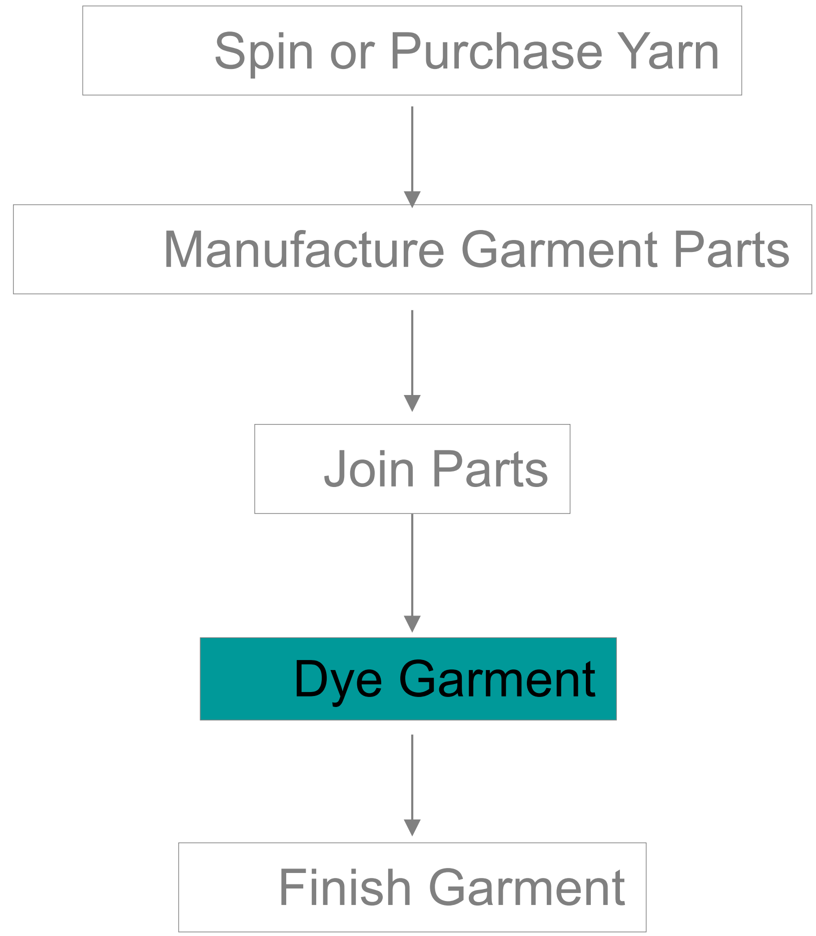
- **Delayed Differentiation**
  - Postpone the step that adds variety to the end of the process.

Benetton
(Sweaters)

- **Managing forecasts of different color sweaters**
  - Lead times were long
  - Capacity constraints were tight
  - Till early 70's sweaters were dyed a colour (per forecast) early in the process.
  - Early 70's changed the sequence…



Old Sequence

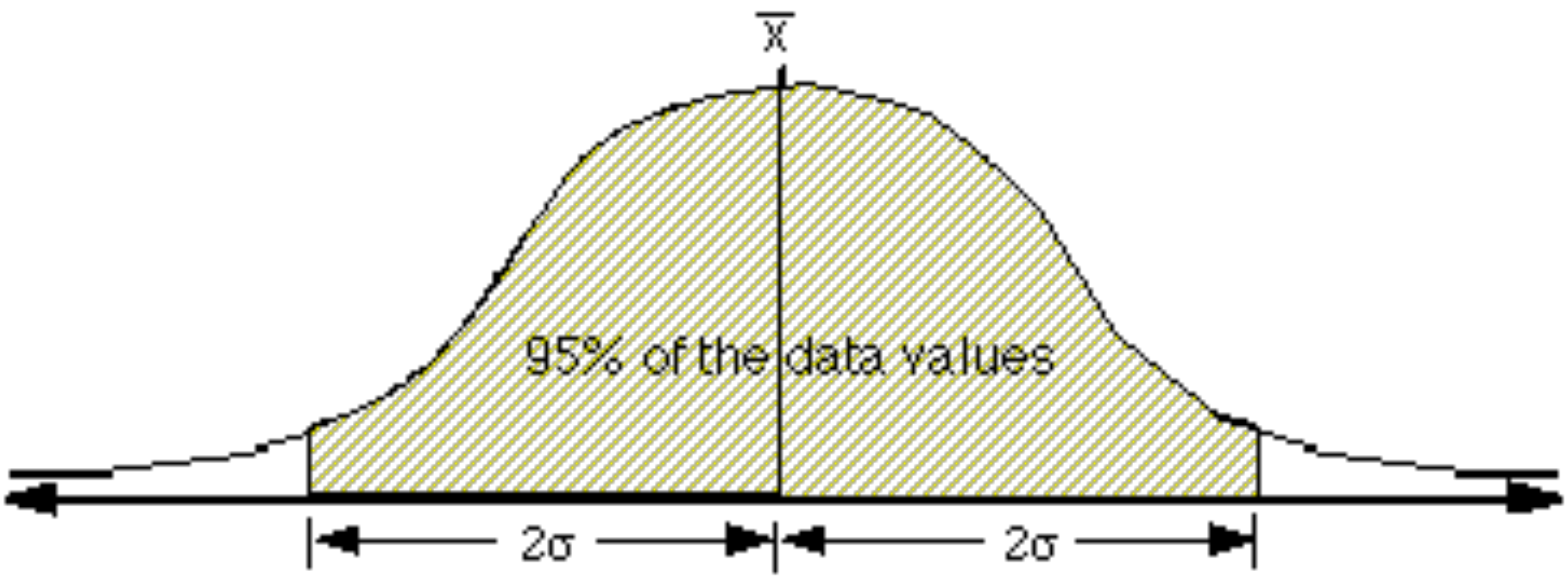


New Sequence

Led to a 10% increase in manufacturing cost but significant decrease in inventory cost and increase in sales

Benetton (Strategy 1)

- Purchase and produce to individual Forecasts separately.
- Let us say Benetton wants a service level of 95% (2 standard deviations from average) for each of its sweaters.


$\overline{x}$
95% of the data values
2σ
2σ

Inventory

- Blue Sweater Inventory:
  - 1000 + 2* 200 = 1400 Units
- Green Sweater Inventory:
  - 2000+2*350 = 2700 Units

- Total = 4100 Units

New Strategy:

(Combine Forecasts For common components and Dye later)

- For the same 95% Service level:
- Inventory will be
  - New Avg = 1000+2000 = 3000

+

  - 2* New Std Dev = 2 * 403

= 3806 Units

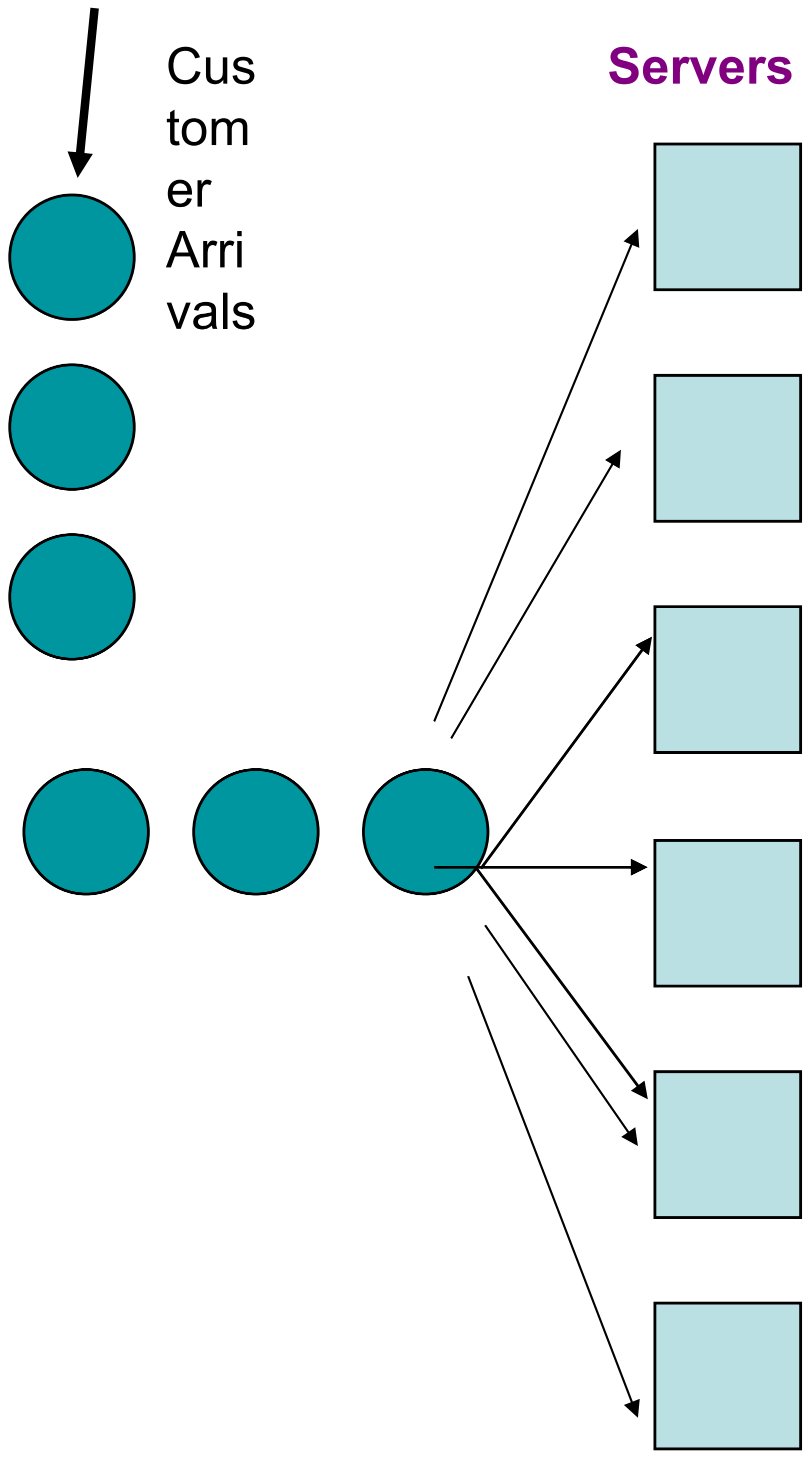
This is exactly Risk Pooling!

- Lowered risk obtained by combining random variables (works many times).

Pooling servers

Single line Multiple servers

- Airports
- Retailers (Non Grocery)



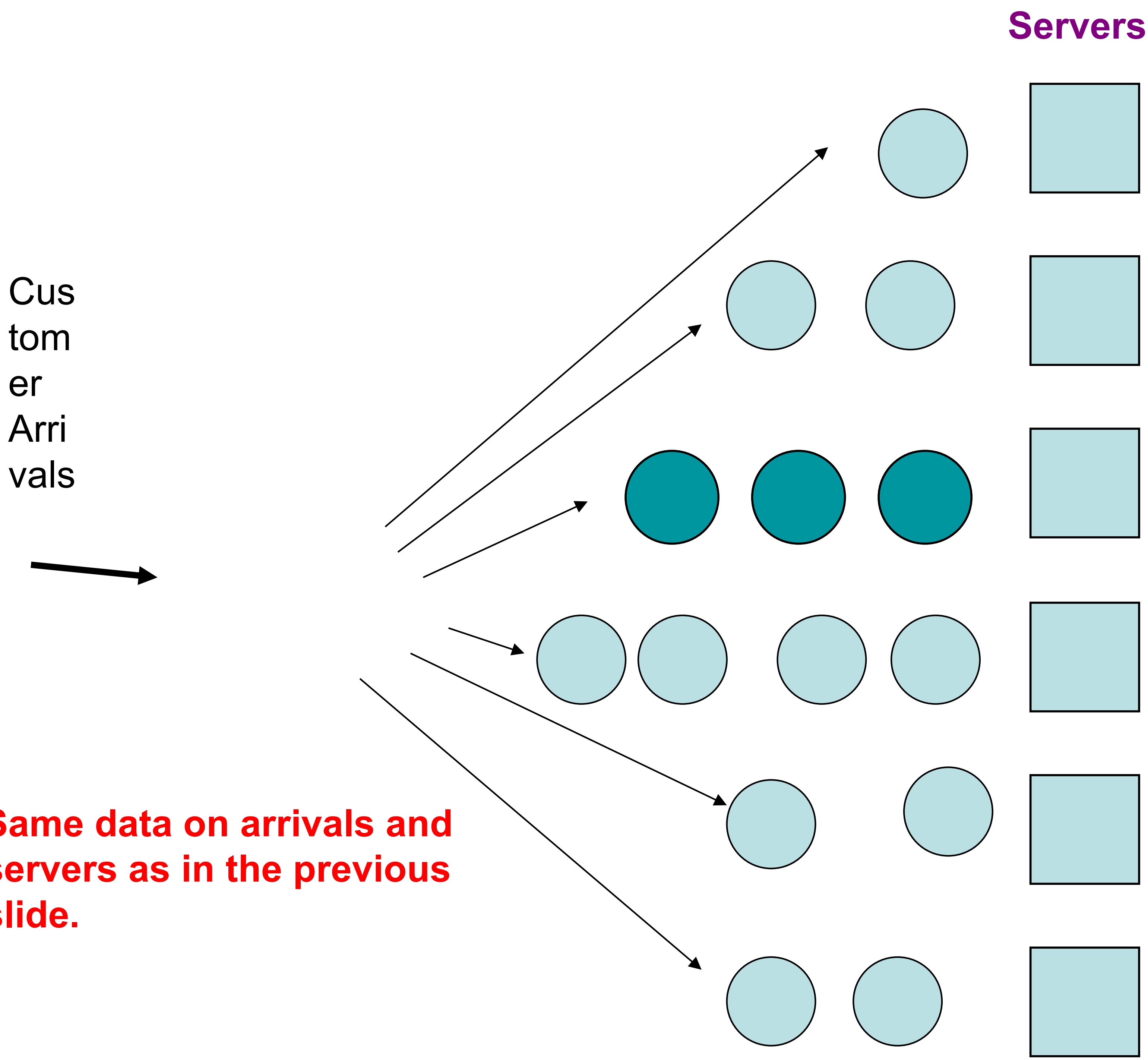
Assume that Jobs or customers arrive at a rate X units per minute.

Assume that each server has a capacity rate / service rate of Y units per minute.

Assume also that arrivals and service times are random but every server is identical.

Multi Lanes-Multiple servers

- Grocery stores
- Movie theatres



Same data on arrivals and servers as in the previous slide.

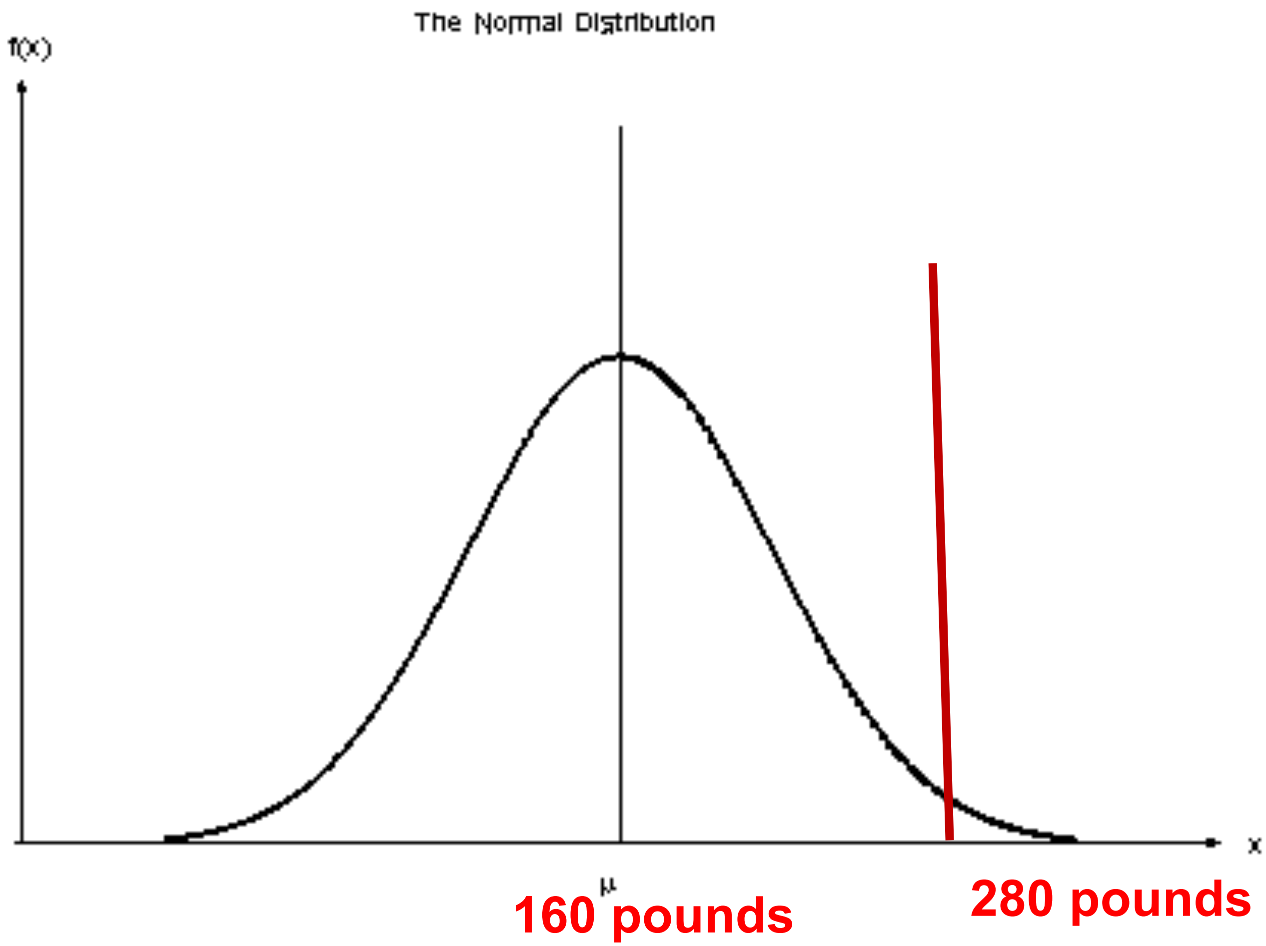
Life Boat Example

Think of passengers in a cruise ship.

Assume (for this example) that weight of passengers is Normally distributed with a mean of 160 lbs and standard deviation of 30 pounds.

Canadian Govt. Safety rules dictate that 99.99% of passengers must have life boats that must hold them.

Weight Distribution



Normal

- 4 standard deviations from the mean approximately covers 99.997% of a normal curve.
  - Therefore choose 160 + 4 * 30 = 280lbs.
  - If we choose individual life boats, we will need to have them be able to stand a weight of 280lbs.

Build 16 person Life boats

- You have to combine 16 normal forecasts each of average 160 and standard deviation = 30.
- The combined forecast will have
  – an average of 16 * 160 = 2560 pounds.
  – Standard deviation of SQRT(16)*30 = 120pounds.

Size of 16 person Life boats

- Again 99.99% safety standard
- Same logic, use 4 standard deviations from the mean:
- Implies the 16 person life boat will have to hold a weight of 2560 + 4*120 = **3040 lbs**.
- If we build 16 individual life boats, we will have to build boats that can hold in total 16*280 = **4480 lbs.**

Other examples…

NBA versus NFL

- Basket Ball
  - Sees Dynasties…
    - Lakers, Bulls, Celtics, Lakers…
- Less in football (if at all…)

NBA versus NFL

Assume team with the higher score on average per possession is the better team.

- Possessions
  - NBA – averages around 90 a game
  - NFL --- averages 10-12 a game.
  - NBA 7 game round robin in play –offs
  - NFL 1 off games.

More Risk Pooling examples…

- Component Standardization
  - Design for manufacture
  - Nokia versus Motorola
  - Black and Decker
- Postponement strategies
  - HP Deskjet
  - Benetton

More Risk Pooling examples…

- Component Standardization
  - Design for manufacture
    - Minimize and simplify components during design process
  - Examples:
    - Nokia as compared to Motorola
    - Black & Decker started in early 70s with uncoordinated set of tools with different motors, housings etc…
    - Developed "Universal Motor" with fixed axial diameter and length that could be adjusted for different power output.

More Risk Pooling examples…

- Postponement strategies
  - HP Deskjet from the mid 80's…Deskjet printers bound for Europe were customized at the plant for their country of destination. Labeling, instructions and, most importantly, the power supply were tailored to the language and electrical conventions of each country.

More Risk Pooling examples…

- Postponement strategies
  - To take advantage of the pooling principle, Hewlett Packard changed their production distribution process by (a) making instructions and labeling generic (i.e., multilingual) and (b) *postponing* installation of the power supply to the distribution center. This allowed them to ship generic European printers to the distribution center and have them customized to a specific country only when orders were received.

    Hence, the forecast only had to be accurate in the aggregate amount, not in the amounts for each country.

Managing Risk (Operationally)

- Risk pooling
  - Why does it work
  - How to implement
    - Joint forecast, common components, Warehouse, delay differentiation, modular design,
- Other things u can do:
    - pre book, produce close to demand (if you can) …

What else can you do to Mitigate the effect of Risk?

- **Think of approaches from different business disciplines…**

Managing Risk

- **OM Triangle:**
  - Build Capacity or Inventory.
- **Finance**
  - Swaps, derivatives, futures…
    - How do you price these instruments?
- **Accounting**
  - Contingencies (Recall Intel)
  - Principal agent models
- **Other strategies**
  - Business Insurance
  - Diversify supplier base

More risk Pooling

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